PORTFOLIO

Faezeh Nikmandan
BIM Specialist | Junior Architect

I'm Faezeh Nikmandan, an architecture graduate (MSc, Politecnico di Torino) who connects creative design with data. With a solid foundation in Building Information Modeling (BIM), I use Revit and AutoCAD to build accurate, data-rich models and coordinated project documentation, and Dynamo and Python to automate repetitive checks and sustainability assessments.
I see BIM as a collaborative process that improves design precision, project efficiency and sustainability. This portfolio shows how I work across scales, from a seven-storey student residence in Collegno to an urban greening programme across six sites and a tourism masterplan in Oman. For each project I describe my role, tools and results.
I'm committed to continuous learning and aim to contribute to multidisciplinary teams with clarity, innovation and purpose

Faezeh Nikmandan

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faezeh-nikmandan

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Education

Politecnico di Torino ,Torino, Italy
- Master of Sciences in Architecture for sustainability 2021-2024

Seraj University , Tabriz ,Iran
- Bachelor in Architecture Engineering 2015-2019

Work Experience

2MIX Studio : Turin, Italy , On-site , Full-time
- Junior Architect : Jan 2026 - Present

Tangram Studio
(Alice e Giorgio Giani Architetti) : Turin, Italy , On-site , Full-time
- BIM Modeler and Junior Architect: Nov 2024 - Sep 2025

Gas Architettura : Turin, Italy , On-site ,Part-time
- Junior Architect : Aug 2023 - Mar 2024
- Curriculum Internship : Jun 2023 - Aug 2023

Scale Carpentry : Iran , Hybrid , Full-time
- Interior Designer : 2020 - 2021

Skills

Language
- English : Advanced
- Italian : Elementary
- Germany : Elementary
- Persian :Native
- Turkish: Advanced
- Azerbaijan : Native

Software Proficiency
- Revit ●●●●○
- Dynamo ●●●○○
- Rhino ●●○○○
- Grasshopper ●●○○○
- DesignBuilder ●●●○○
- 3D Max ●●●●○
- AutoCad ●●●●●
- Lumion ●●●○○
- V-Ray ●●●○○
- Enscape ●●●●○
- Unreal Engine ●●●○○
- Photoshop ●●●●○
- InDesign ●●●●○
- Illustrator ●●●●○
- Excel ●●●○○
- PowerPoint ●●●●●
- Word ●●●●●

Programming Language
- Python ●●○○○
- Html ●●●○○
- Css ●●●○○
- JavaScript ●●○○○
- React ●●○○○

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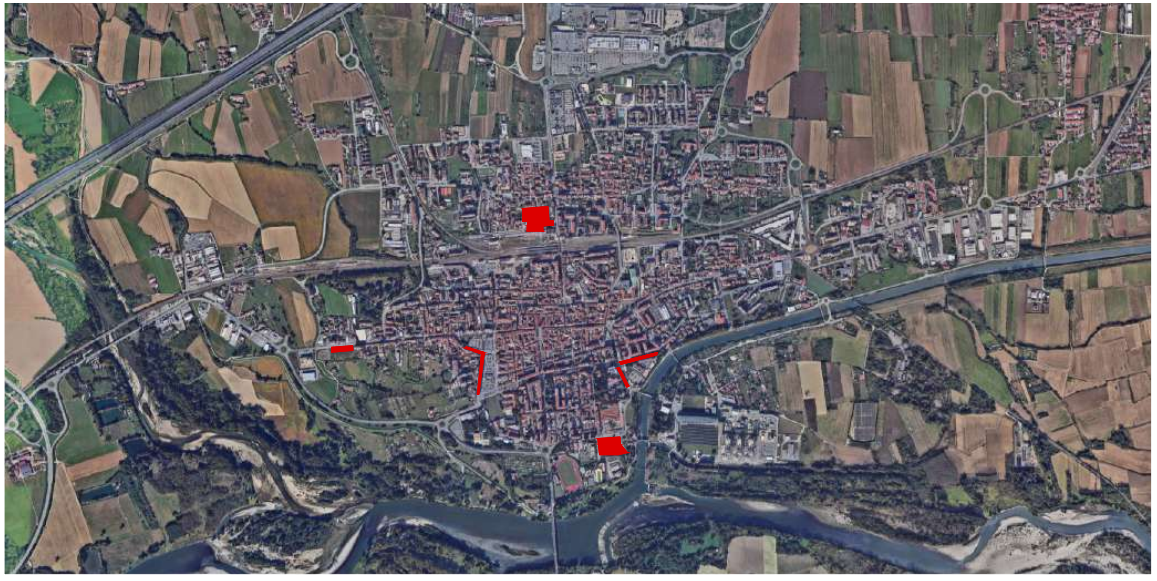
Professional projects – Landscape and Urban design

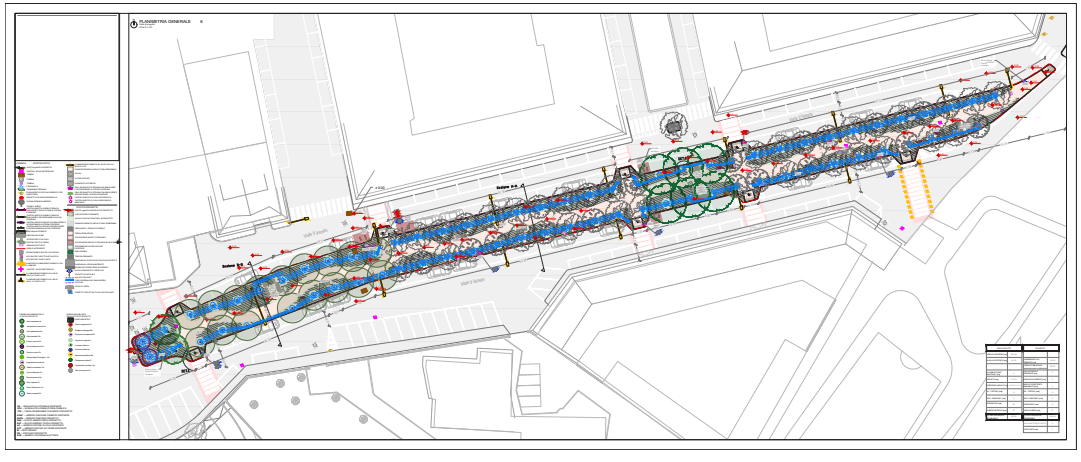
Year :2026
Studio : 2MIX
Location : Turin Metropolitan, Italy

An urban greening and public-space regeneration project spanning six intervention areas. The project transforms existing streets and parking areas into greener, more accessible spaces that respond to the needs of pedestrians, drivers and local residents.

Developed from the approved PFTE, the executive design integrates new planting, depaving, permeable surfaces and sustainable rainwater management with the reorganisation of parking and circulation.
Detailed proposals address parking capacity and bay arrangements, vehicle manoeuvring space, continuous pedestrian routes and crossings designed to improve visibility. Green areas, seating, waste bins and traffic signage are coordinated within each site, balancing everyday functionality with shade, comfort and environmental quality.

Together, the six interventions aim to reduce urban heat, increase soil permeability and strengthen the city's green infrastructure while improving the usability and character of its public spaces.

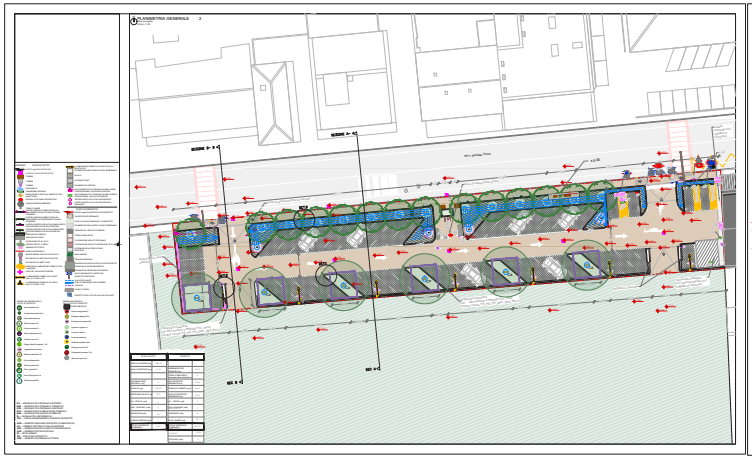


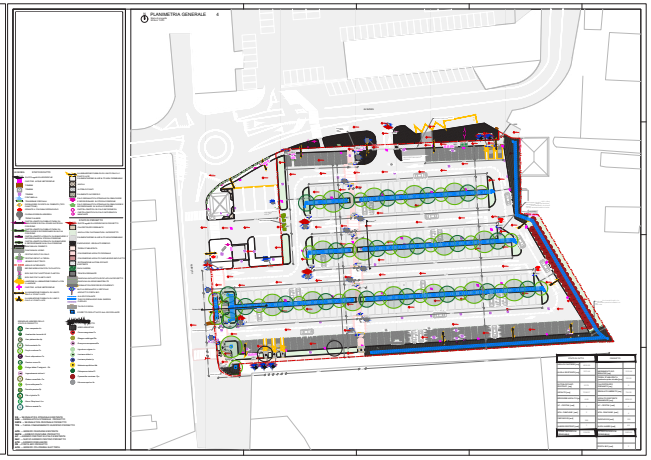


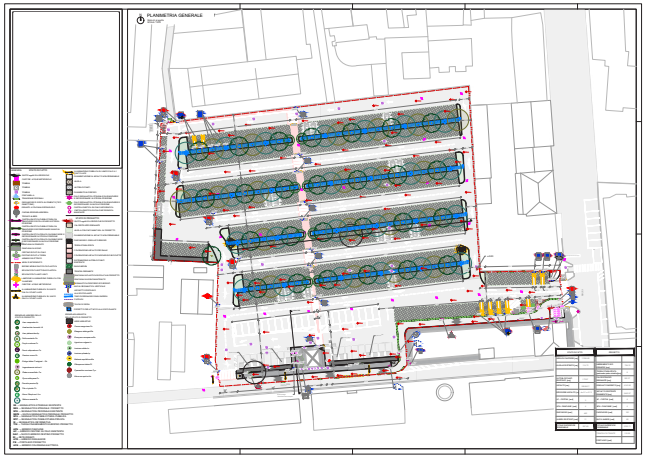
My role : Design development, executive drawings & cad coordination

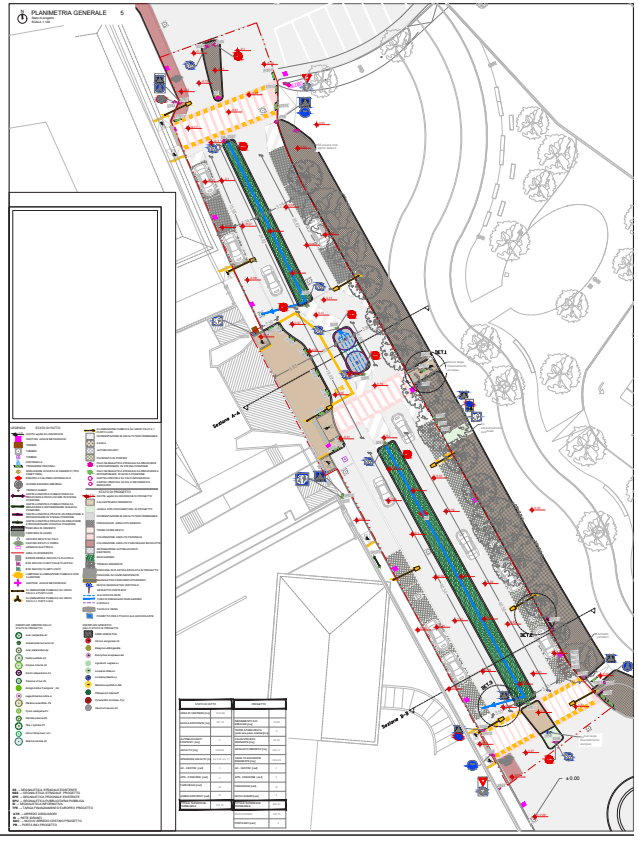
From a design point of view, my role was to transform the existing urban environments across all six intervention areas into more sustainable and efficient spaces for pedestrians, vehicles and local residents. Building on the approved PFTE, I developed parking layouts with reference to Italy's Codice della Strada, together with green areas, urban furniture and traffic signs.

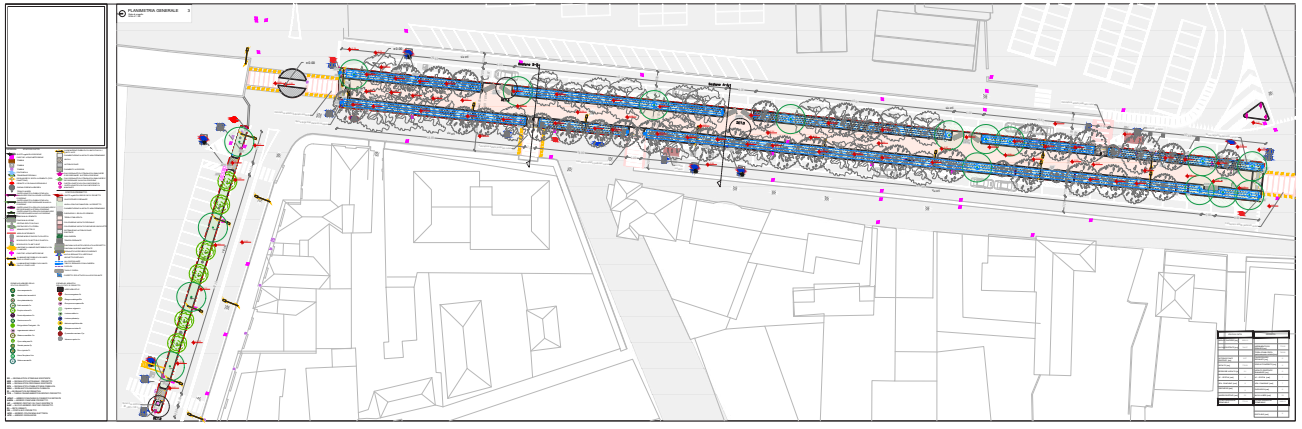
I determined how many cars could fit in each parking area, how the parking spaces should be arranged, and the street and manoeuvring widths required for perpendicular, parallel or 45-degree parking. I also decided where











to place benches and trash bins, and designed pedestrian paths and green areas.

Particular attention was given to pedestrians crossing the street. I designed pavement extensions aligned with the outer edge of the roadside parking spaces, allowing pedestrians to move beyond parked cars and see approaching traffic more clearly, while also making them more visible to drivers.

My drawing responsibilities included preparing the rilievo and stato di fatto drawings, as well as executive plans and sections. I also organised the AutoCAD workspace using external references (Xrefs), enabling five team members to work simultaneously on different drawings within the same project.

Professional projects – Student Residence Hall

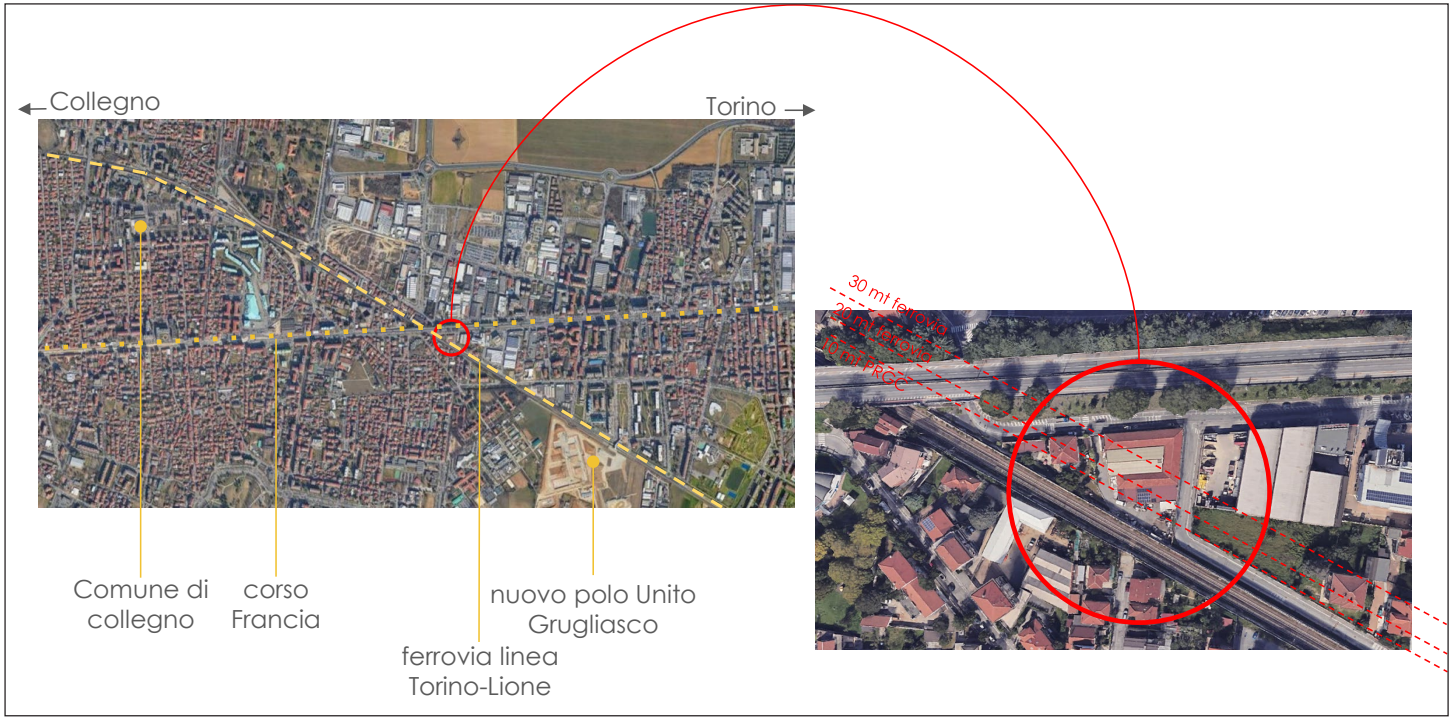
Year :2024-2025
Studio :Tangram
Location : Comune di Collegno , Italy

The project involves the construction of a new seven-storey student residence on Via Torino, near Corso Francia, in Collegno, Turin.

Replacing existing structures, the development combines student accommodation with shared spaces and service areas in a well-connected urban setting.

My role:

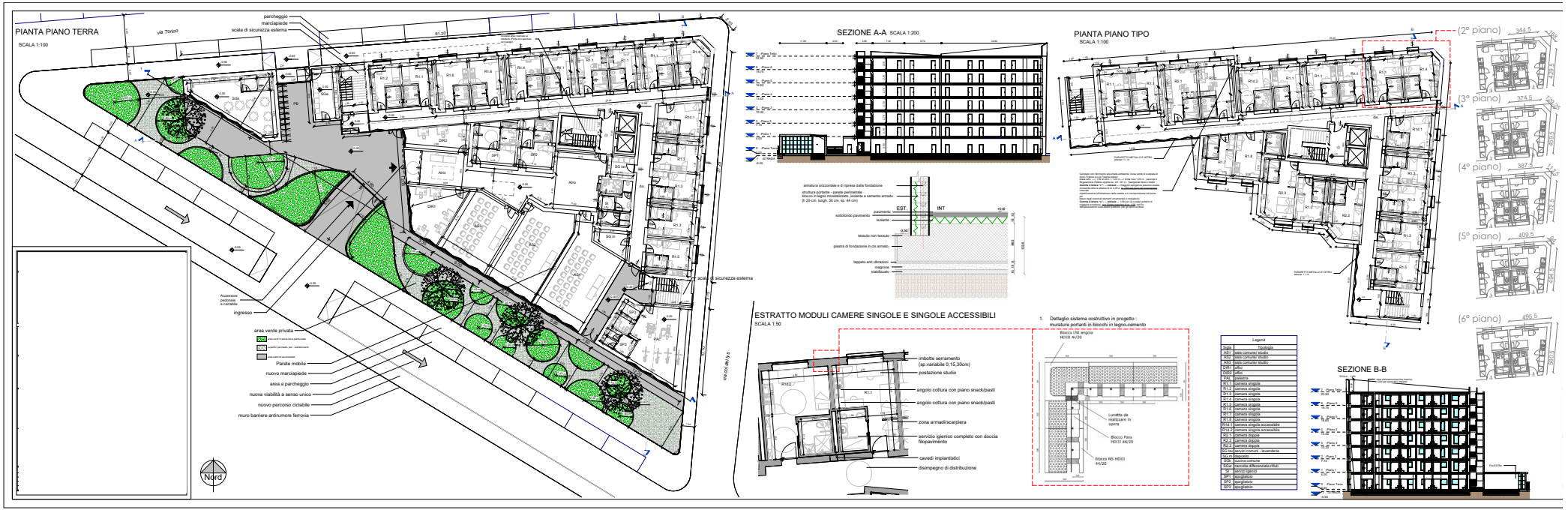
I contributed to architectural development across different project phases, creating and updating detailed BIM models and coordinating existing-condition information with design proposals.



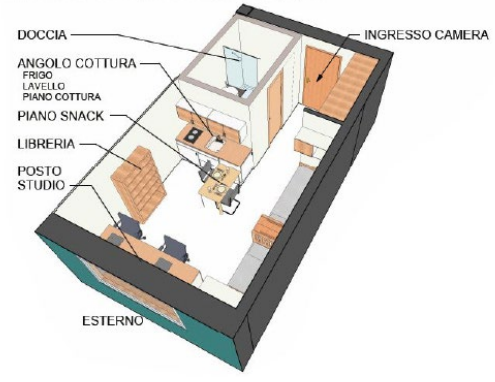
All project documentation and sheet layouts were produced in Autodesk Revit, including plans, sections, elevations and comparative plans (red and yellow drawings) identifying proposed construction and demolition works.

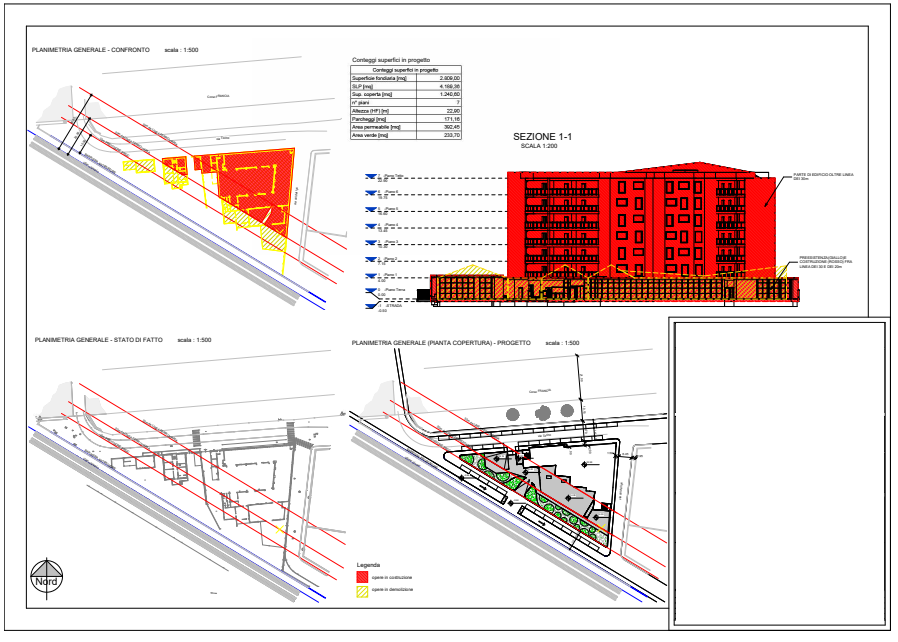
I also prepared Revit-based quantity take-offs and schedules for building elements and spaces, maintaining consistency between the model, drawings and documentation.

I developed 3D renderings directly in Revit and prepared presentation sheets to communicate the proposed architecture and spatial organisation.



vista interna di una camera doppia





Professional projects – International Urban Design

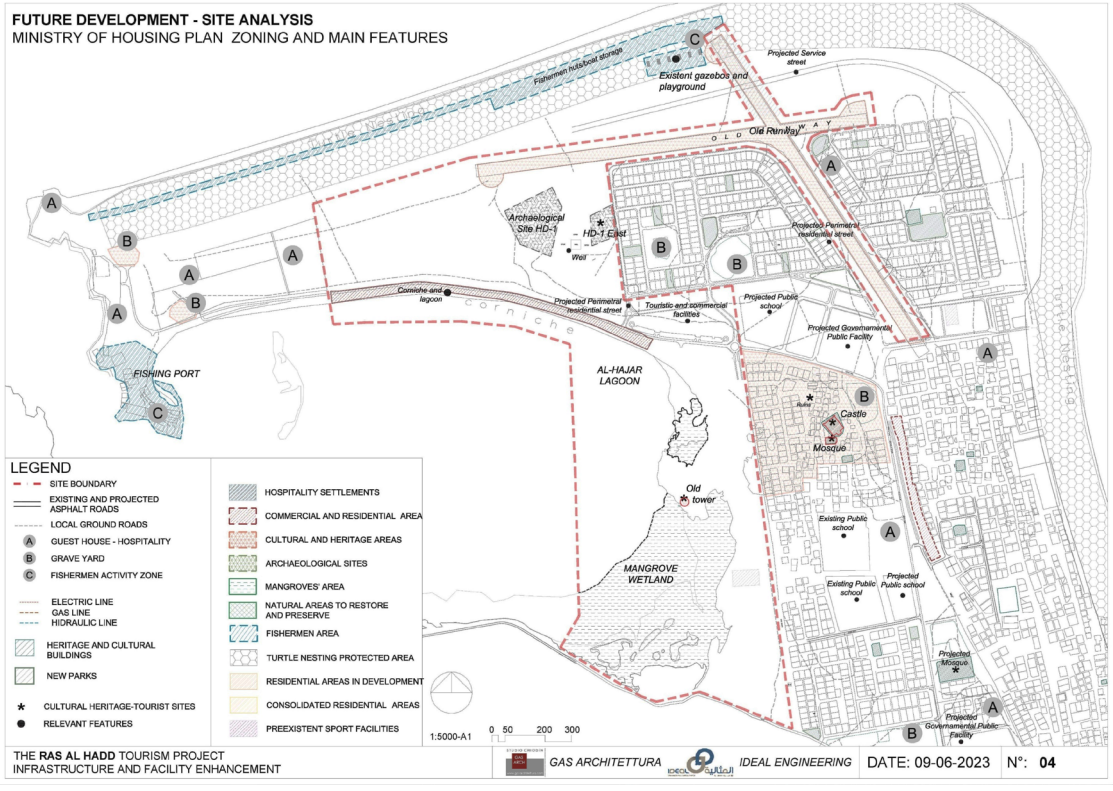
Year :2023-2024

Studio : Gas Architettura -www.gasarchitettura.com

Location : South Al Sharqiyah Governorate, Oman

This project aimed to enhance tourism facilities in Ras Al Hadd to boost both domestic and international tourism through sustainable development. It focused on preserving the natural environment, including turtle nesting beaches, lagoon, and wetlands, while integrating cultural and historical assets like the fort, tower, and old runway.

The project also sought to create opportunities for public and private investments and support local businesses through innovative improvements. By emphasizing ecological and cultural preservation, the project aimed to increase visitor numbers and create lasting positive impacts for the local community and the region's tourism sector.



As an architectural assistant at Gas Architettura Studio, I collaborated with the team on both the conceptual and detailed design stages, focusing on sustainable tourism infrastructure. Together, I conducted extensive research on the area's natural and cultural heritage, which informed our design decisions to balance development and conservation. Visual representations and presentations were created by me along with the team to effectively communicate design ideas to clients and stakeholders, making it easier for them to understand and approve the project plans. Additionally, detailed architectural drawings and documentation were prepared to ensure the project's execution and compliance with local regulations.

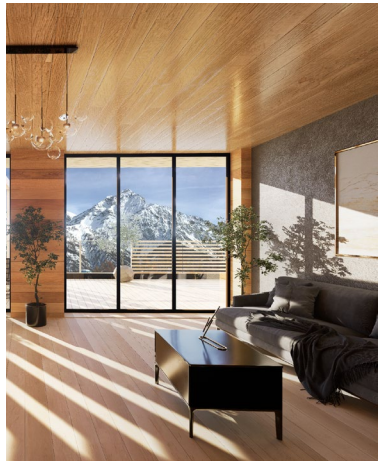
Professional projects - Residential Project

Year :2023-2024

Studio : Gas Architettura -www.sunsicario.com , www.gasarchitettura.com

Location : Sansicario Torinese,Italy

The project for the new Village Ski Hotel Resort CLUB MED in Sansicario is located in the municipality of Cesana Torinese, in the San Sicario Alto area, within the Via Lattea ski area. It includes a new hotel structure with 1,000 beds, a wellness center, a spa, swimming pools, and restaurants serving local specialties.



As an architectural assistant at Gas Architettura Studio, my responsibilities included:

- Creating 2D drawings, such as executive detailed plans,elevations and sections, using AutoCAD.
- Developing a detailed 3D model in Revit, including elevations, plans, and sections, as well as producing interior and exterior renders using Enscape.
- Producing isometric plans and incorporating proposed materials using Photoshop.

Academic project- Master thesis - BIM and LEED Integration for Water Conservation

Year :2024
Location : Politecnico di Torino

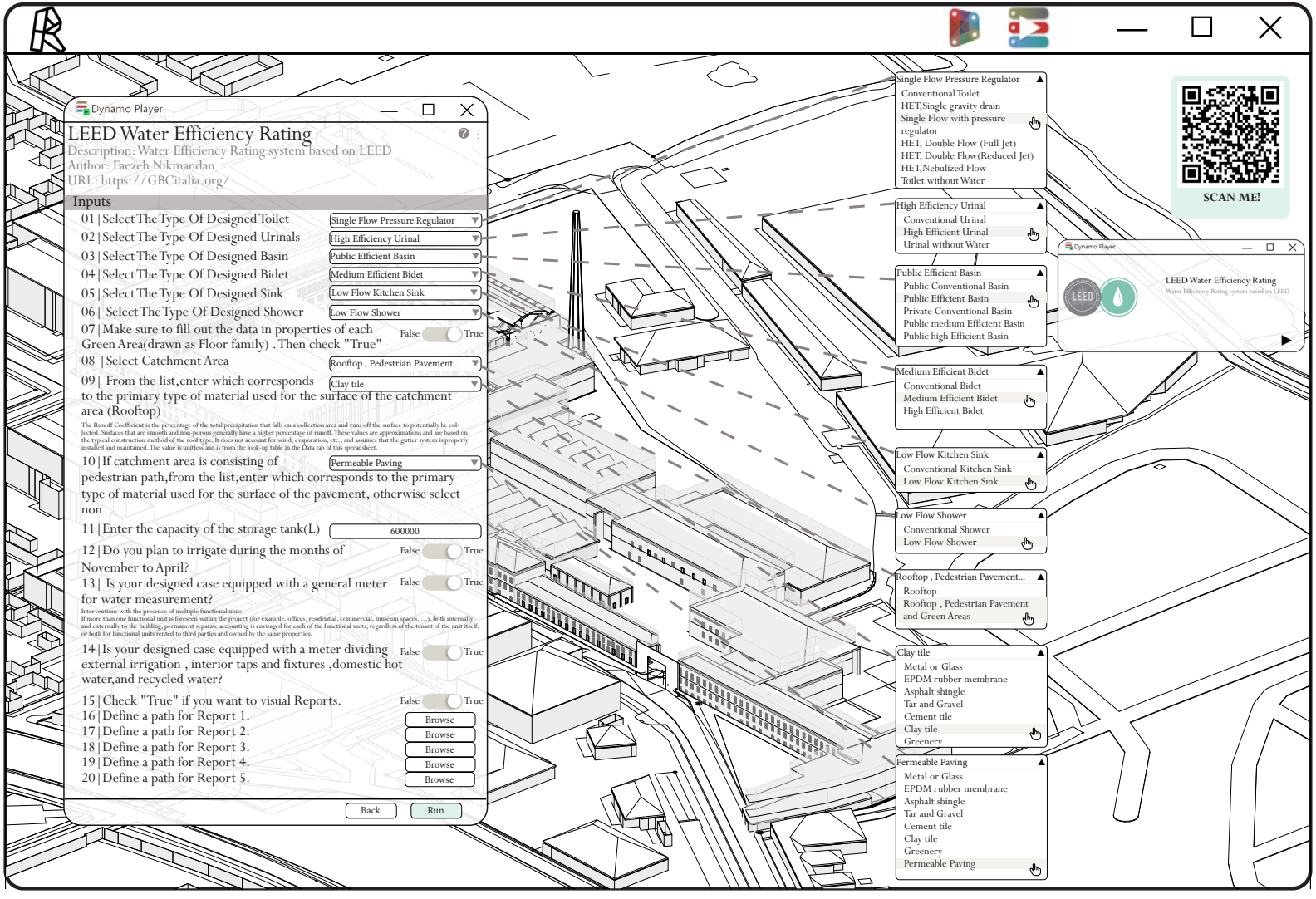
The thesis examines the green building concept with a focus on water conservation, using Building Information Modeling (BIM) within the LEED Water Efficiency framework.

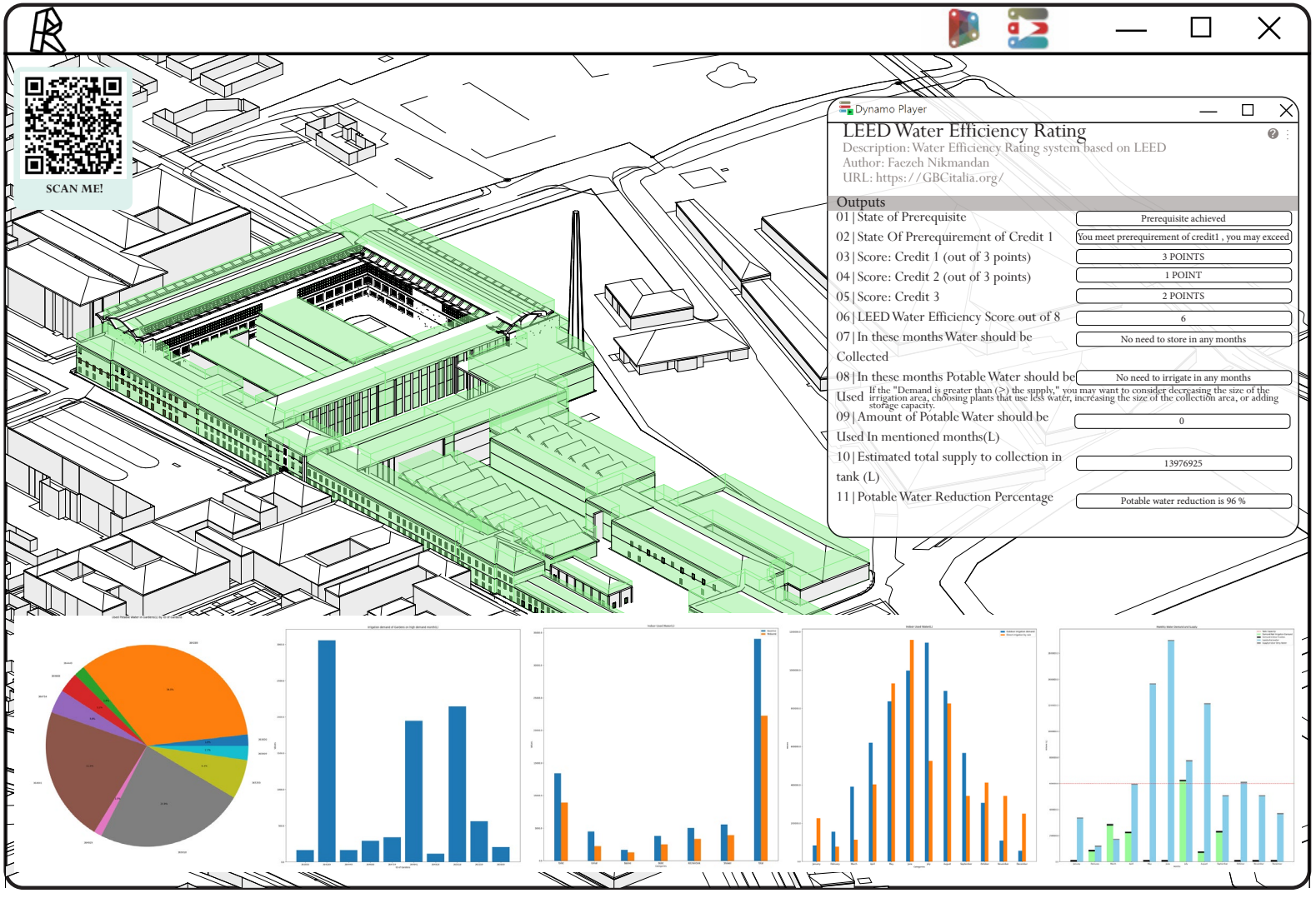
I developed a methodology combining Autodesk Revit, Dynamo visual programming and Python scripting within selected Dynamo nodes to automate assessment calculations and help users select appropriate water-saving strategies and pursue relevant LEED credits.

Applied to Ex Manifattura Tabacchi in Torino, the methodology supports the integration of potable water conservation during the early stages of design.

A conceptual model developed in Revit provides the geometric information needed to assess different indoor water-consuming fixtures and outdoor landscaping choices.

The Dynamo workflow, supported by Python scripts, connects model information with assessment inputs to calculate indoor and outdoor water demand, evaluate rainwater availability and storage adequacy, and estimate potable water reduction percentages. Through Dynamo Player,





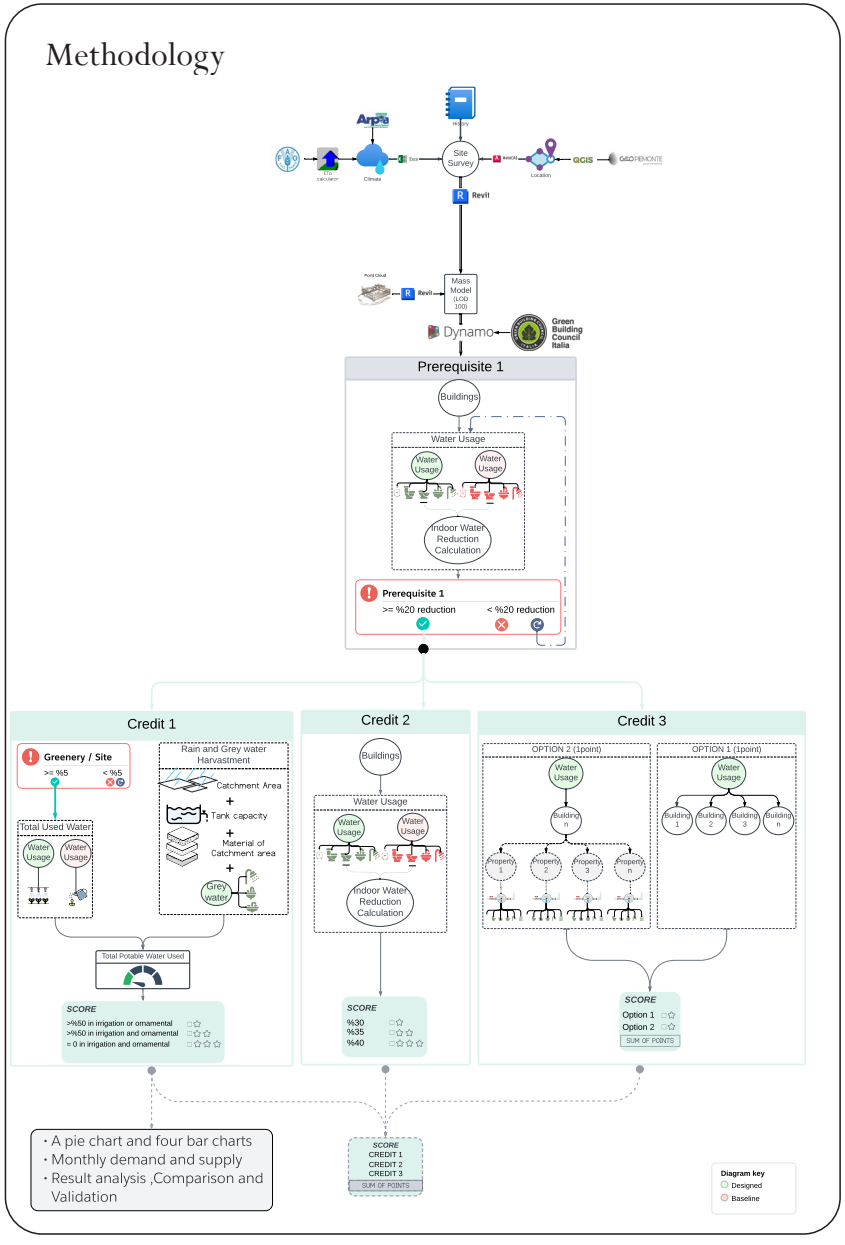
users can adjust parameters and compare alternatives without modifying the underlying code.

The methodology was tested through three scenarios representing low, medium and high water demand. The comparisons demonstrate how fixture selection, planting choices and irrigation strategies influence water consumption and the fulfilment of selected LEED prerequisites and credits. Requiring only a low-detail conceptual model and readily available inputs, the workflow supports assessment before detailed design.

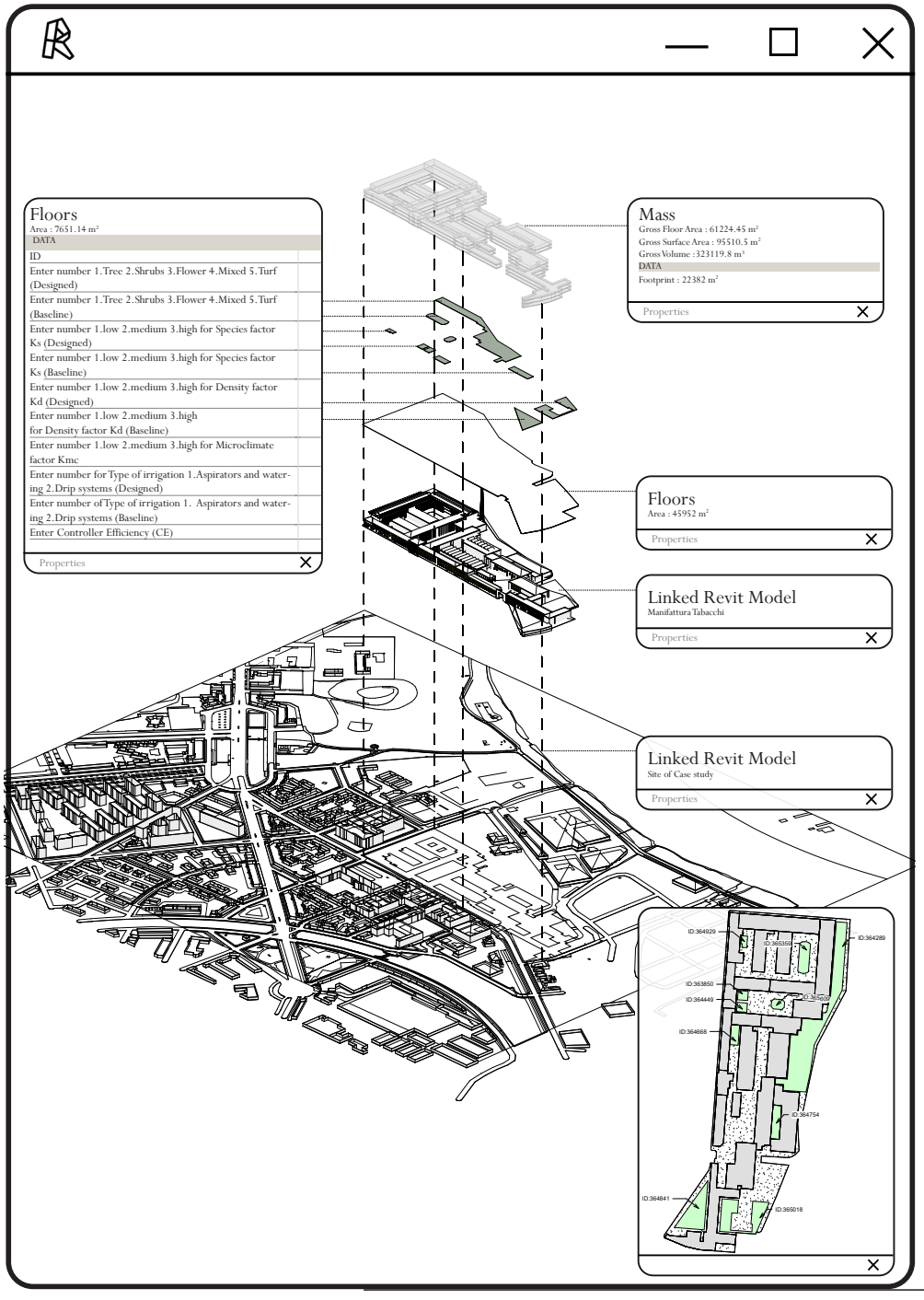
Five graphical outputs, accompanied by explanatory text, communicate irrigation requirements, monthly water balances, storage adequacy, potable water savings and selected LEED Water Efficiency credit scores.

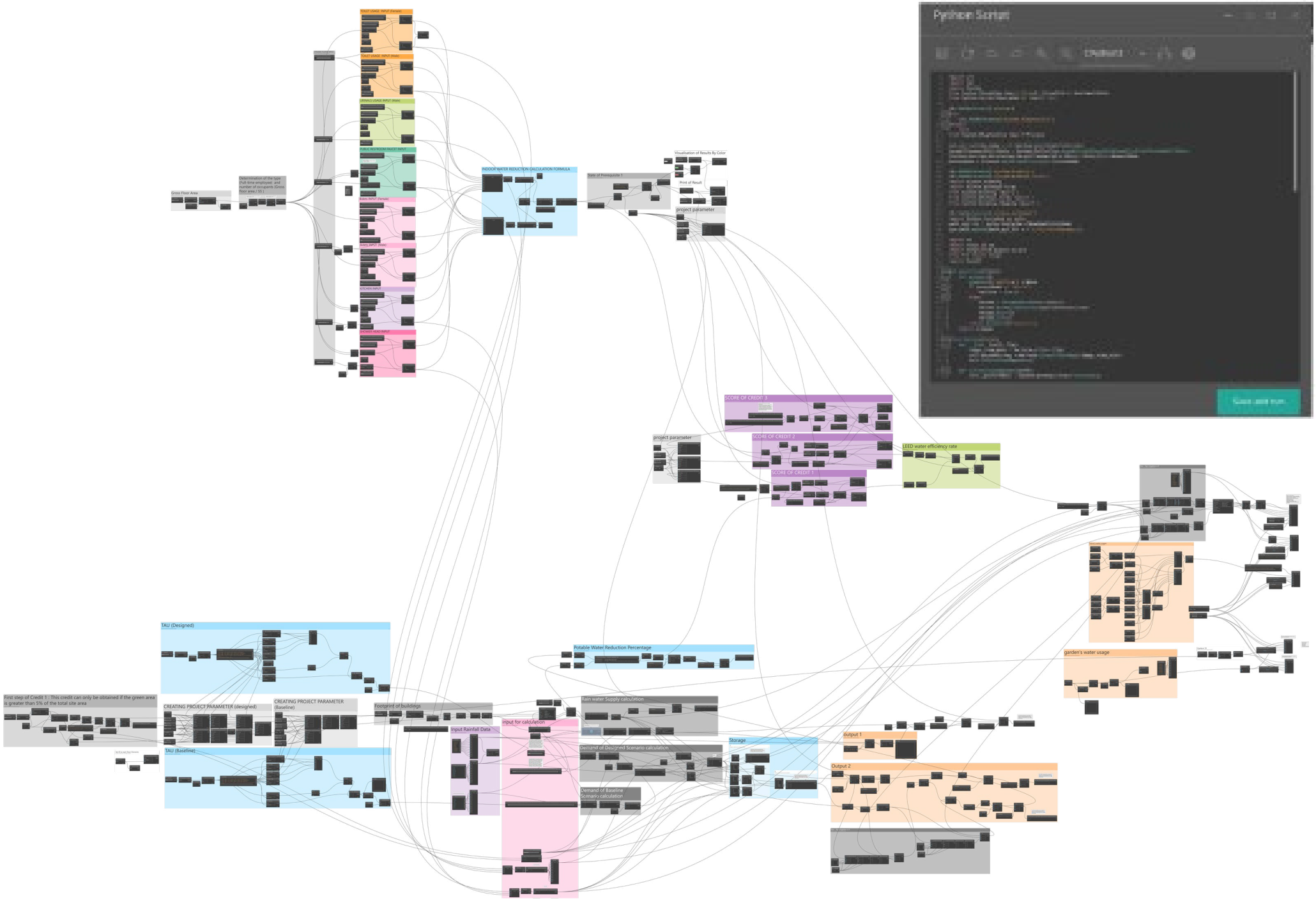
These results help architects, landscape designers and other stakeholders make informed decisions about water-efficient projects.

Methodology



While the thesis focuses on water efficiency, it also identifies opportunities to extend automation to other LEED categories. Full LEED certification requires assessment across the wider certification framework.




Gross Floor Area
Determination of the type (Full-time employee) and number of occupants (Gross floor area / 55)
TOILET USAGE INPUT (Female)
TOILET USAGE INPUT (Male)
URINALS USAGE INPUT (Male)
PUBLIC RESTROOM (NUCET INPUT
BUBS INPUT (Female)
BUBS INPUT (Male)
KITCHEN INPUT
SHOWER HEAD INPUT
INDOOR WATER REDUCTION CALCULATION FORMULA
State of Prerequisite 1
Visualisation of Results By Color
Print of Result
project parameter
project parameter
SCORE OF CREDIT 3
SCORE OF CREDIT 2
SCORE OF CREDIT 1
LEED water efficiency rate
TAU (Designed)
TAU (Baseline)
First step of Credit 1 - This credit can only be obtained if the green area is greater than 5% of the total site area
CREATING PROJECT PARAMETER (designed)
CREATING PROJECT PARAMETER (Baseline)
Footprint of buildings
input Rainfall Data
input for calculation
Potable Water Reduction Percentage
Rain water Supply calculation
Demand of Designed Scenario calculation
Demand of Baseline Scenario calculation
Storage
output 1
Output 2
garden's water usage

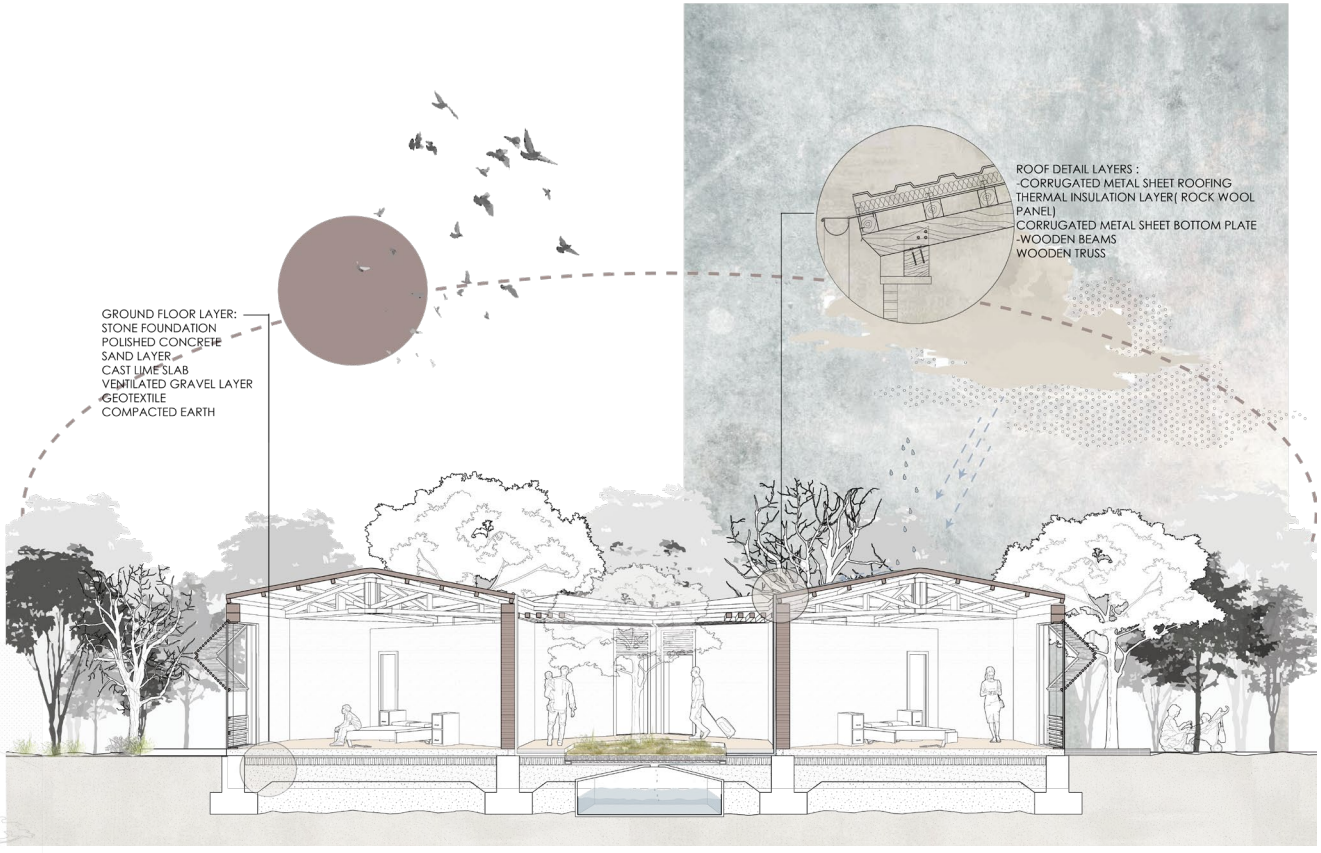
Academic project- Master Degree

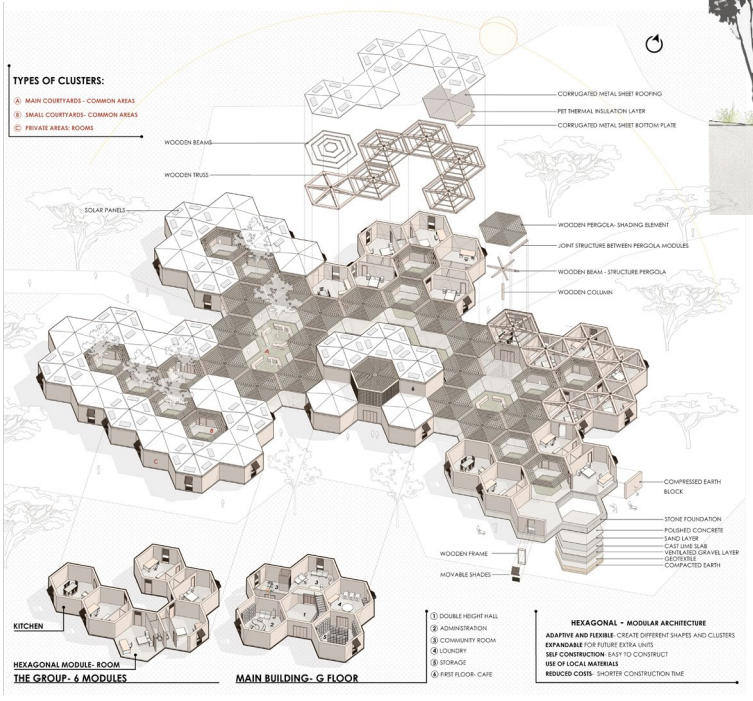
Year :2022
Location : Politecnico di Torino

The project comprises six hexagonal clusters, each containing five rooms and a kitchen, linked by a shaded wooden pergola and a central communal building. The modular design combines shared courtyards, locally sourced materials and opportunities for future expansion. Cross ventilation and shading strategies were assessed using DesignBuilder for Cape Town's climate.

My role:

I produced architectural plans and sections in Autodesk Revit and completed post-production in Adobe Photoshop, refining the drawings and presentation graphics.



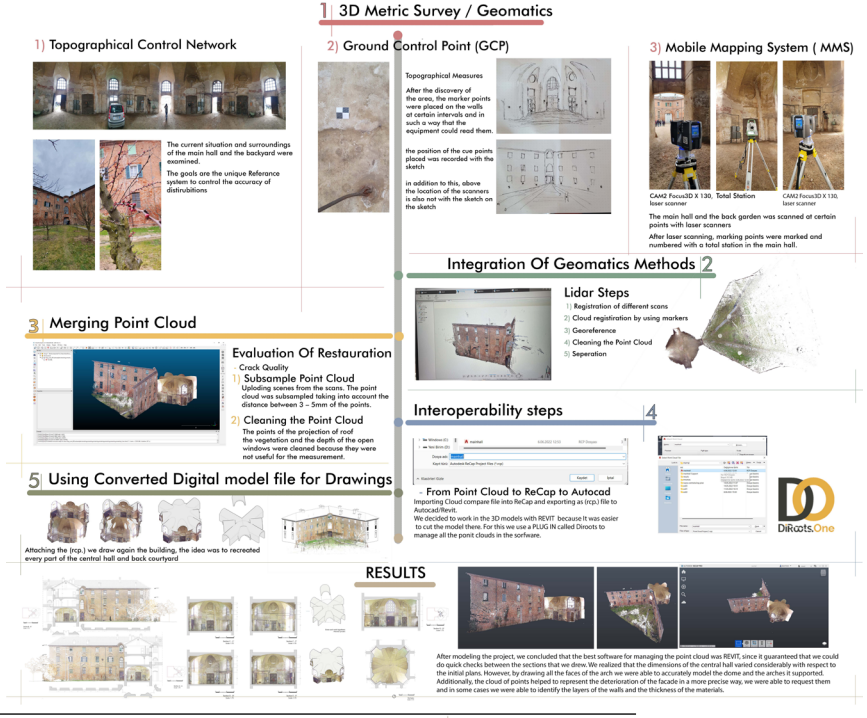


Academic project- Master Degree

Year :2022
Location : Politecnico di Torino

I contributed to the metric survey and digital reconstruction of the Stupinigi Kennels, supporting laser-scanning and topographic survey activities. I processed survey data through point-cloud registration, georeferencing, cleaning and merging in CloudCompare, then prepared the files in Autodesk ReCap for use in Revit.

Using the point cloud as a reference, I reconstructed the central hall and rear courtyard in Revit and produced architectural plans, sections and elevations, documenting the existing geometry and supporting the restoration analysis.

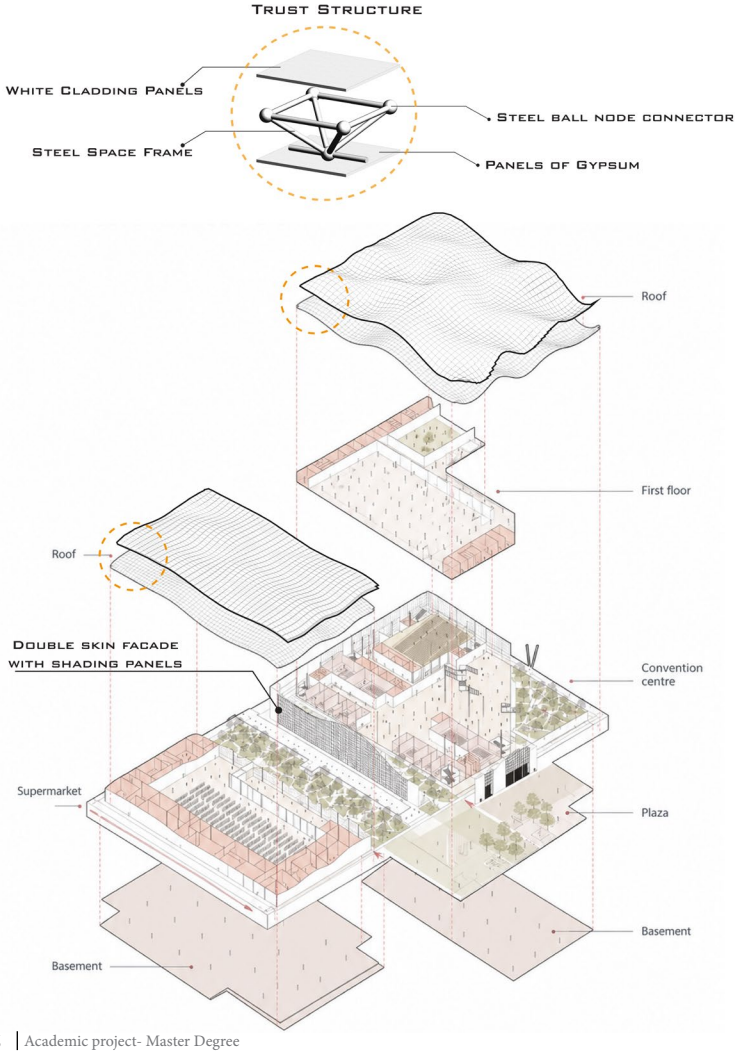


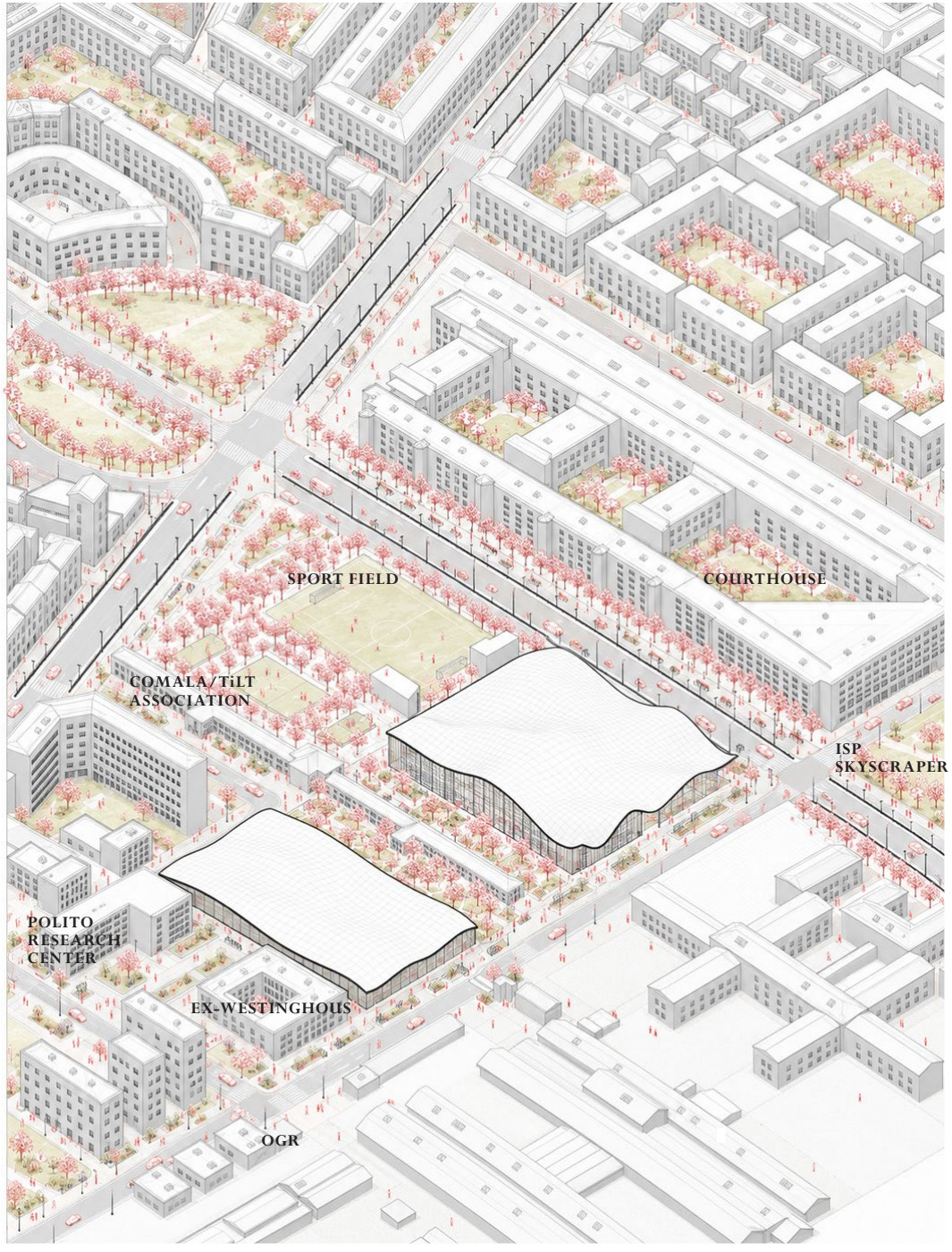
Academic project- Master Degree

Year :2023

Location : Politecnico di Torino

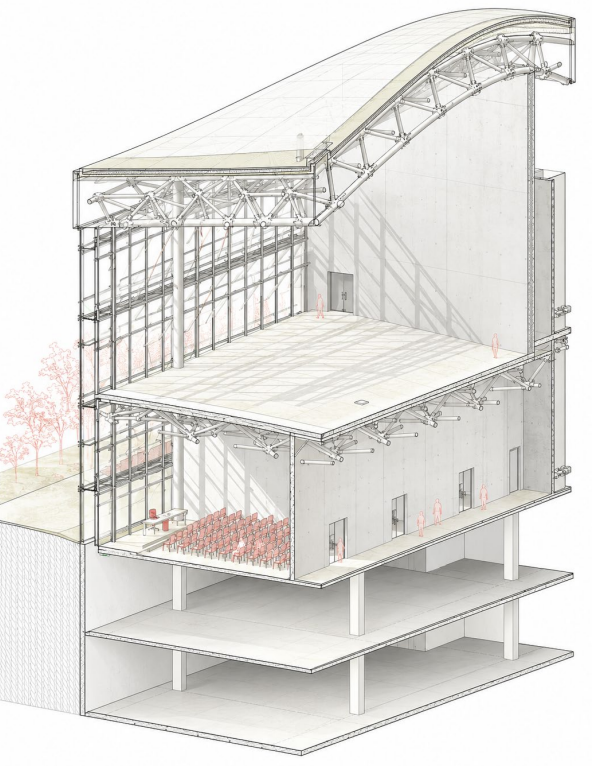
The project combines a convention centre and a supermarket, with an existing historic barracks acting as a transition between the two functions and the centrepiece of a public plaza. A curved roof establishes architectural unity, while clearly defined entrances and underground parking support access. The car-free plaza integrates landscaping to connect the development with nearby green spaces, including Giardino Nicola Grosa and the square on Corso Ferrucci.





BIM MODELLING & PARAMETRIC COMPONENTS:

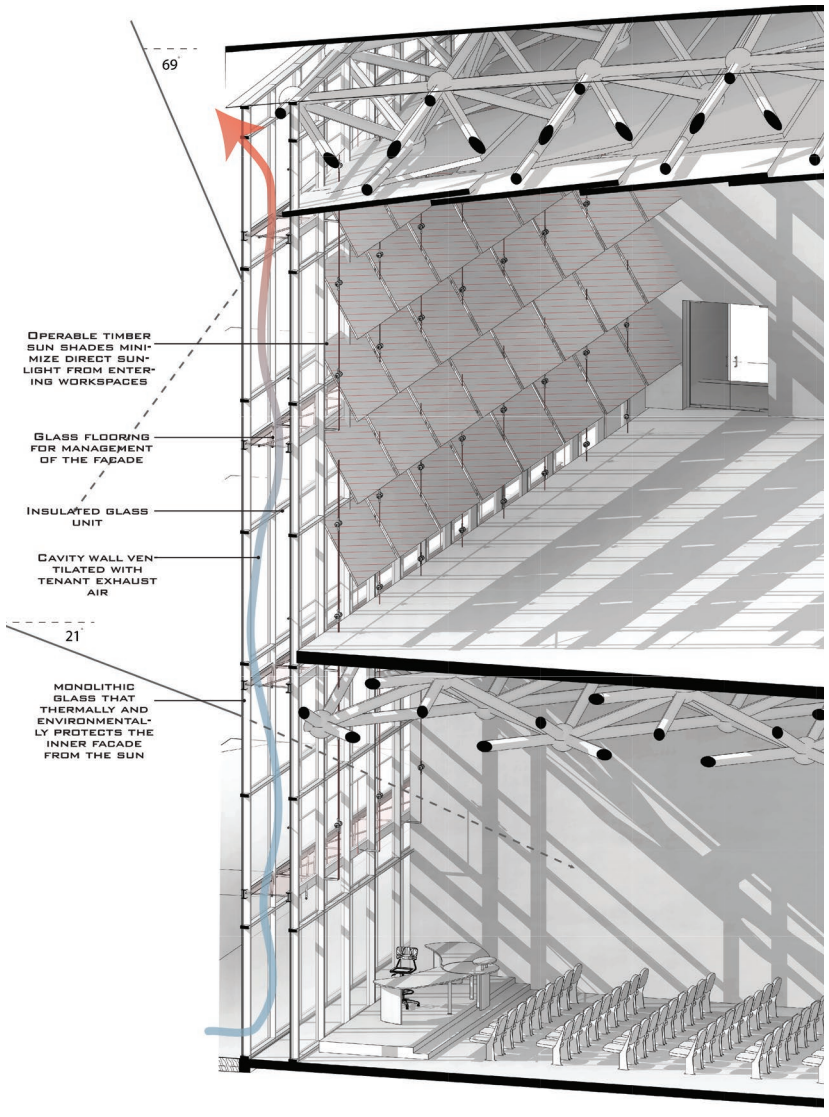
I developed the entire project in Autodesk Revit, including architectural modelling and family-based development of the roof structure and its components. The roof assembly incorporates white cladding panels, a steel space frame, steel ball-node connectors and gypsum panels. A double-skin façade with shading panels forms the building envelope, integrating architectural expression with environmental performance.

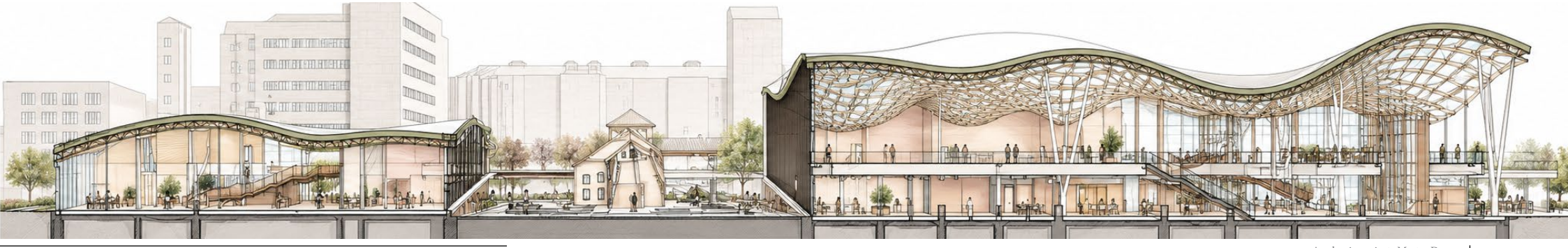


BUILDING PERFORMANCE & DAYLIGHTING:

Using DesignBuilder, I evaluated daylighting, ventilation and thermal performance, supported by illuminance maps and heat maps. My design targets included adequate daylight for visual tasks, reduced dependence on artificial lighting, balanced luminance and glare control. Shading and glazing strategies also aimed to limit direct solar discomfort and summer overheating, reducing cooling demand. The glazing target combined 47% solar-energy reflectance with visible light transmittance above 61%.

The project brought together parametric modelling, technical problem-solving and performance-based design, alongside critical thinking and visual communication, to connect architectural decisions with building performance.





Academic project- Bachelor Degree

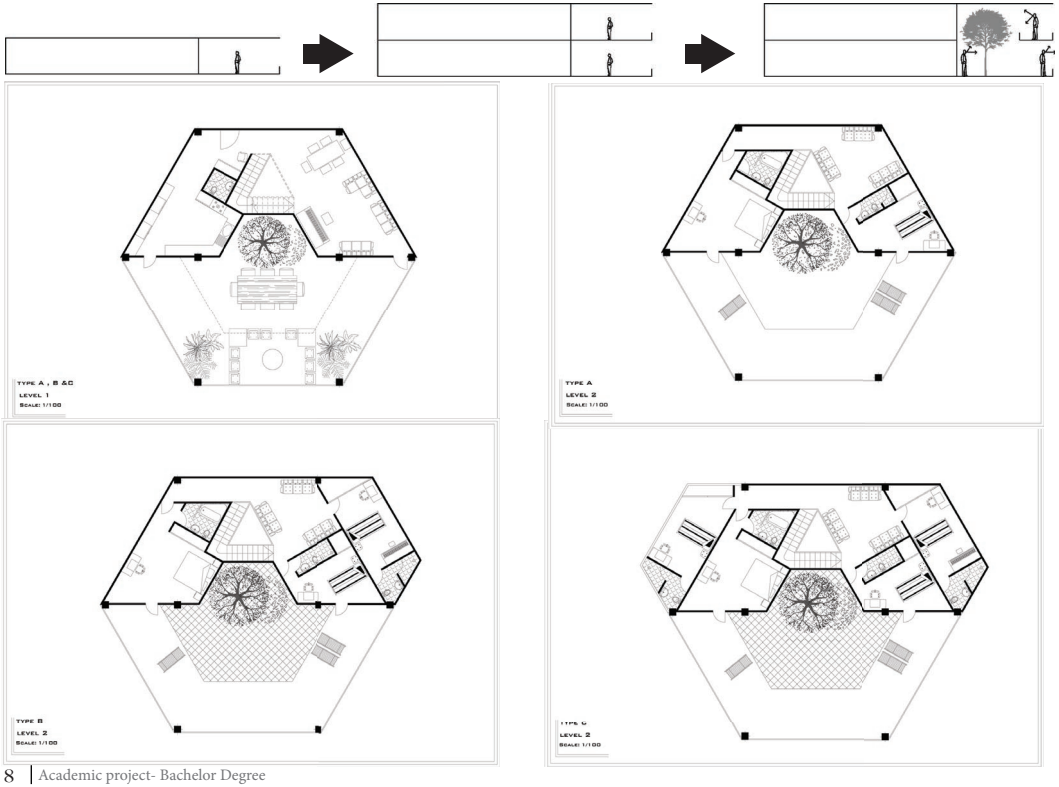
Year :2018
Location : Iran

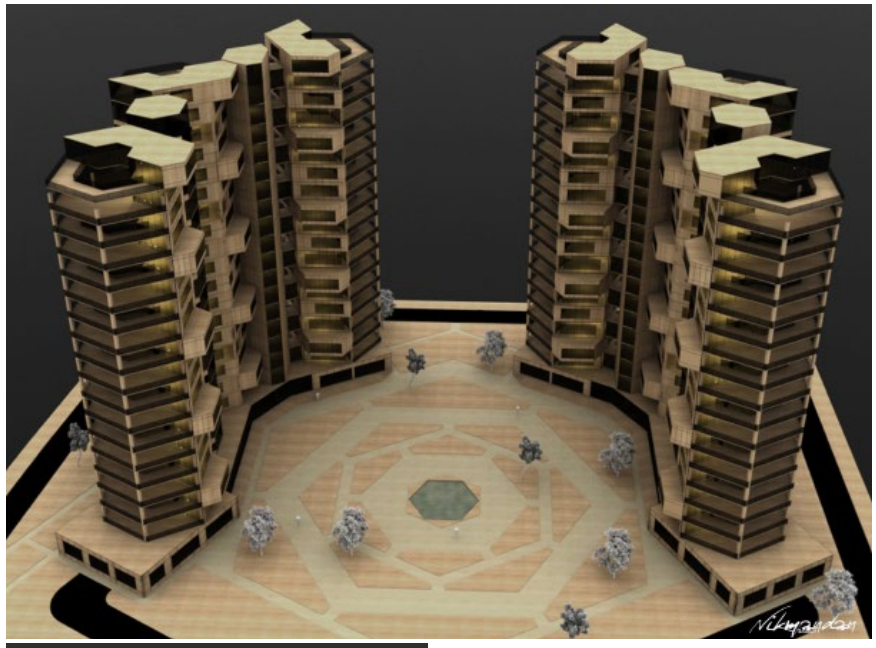
My design strategy focused on residents' needs, reimagining conventional residential blocks with small terraces as two-level homes with expansive terraces. The concept strengthens the connection between indoor living spaces, greenery and open skies.

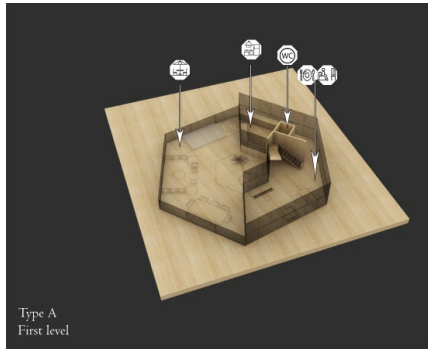
I chose a hexagonal form to create multiple orientations and wider views of the surroundings. The blocks are arranged to maintain direct views of the landscape, while generous terraces extend everyday living outdoors.

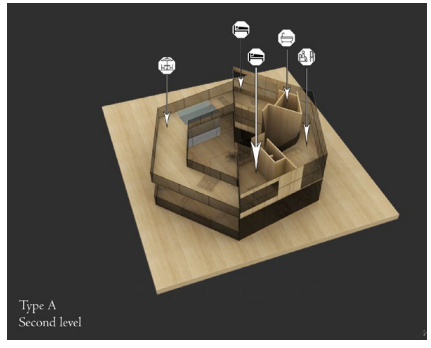
The complex includes three housing types, all sharing the same first-level layout but differing on the second floor: Type A has one bedroom, Type B has two bedrooms, and Type C has three bedrooms, responding to different household needs.

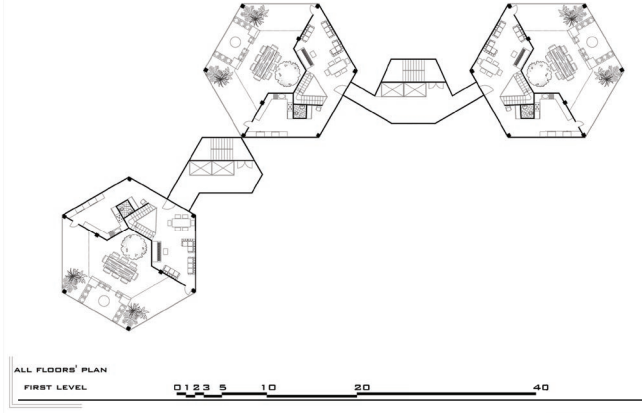
I created the 3D architectural model in Autodesk 3ds Max and produced renderings using V-Ray to communicate the building forms, materials, lighting and relationship with the landscape.





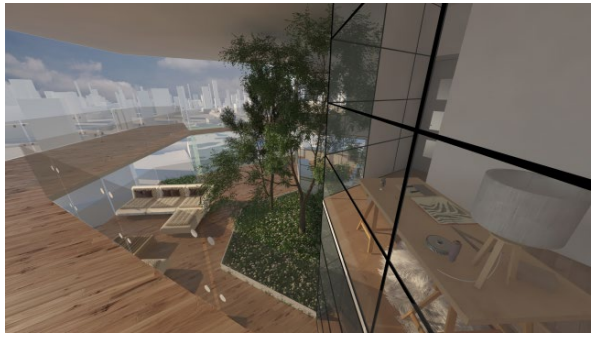






Architectural Visualizations- Bachelor Degree

Year :2018
Location : Iran



Thank You

"I don't believe architecture has to speak too much. It should remain silent and let nature in the guise of sunlight and wind."

TADAO ANDO